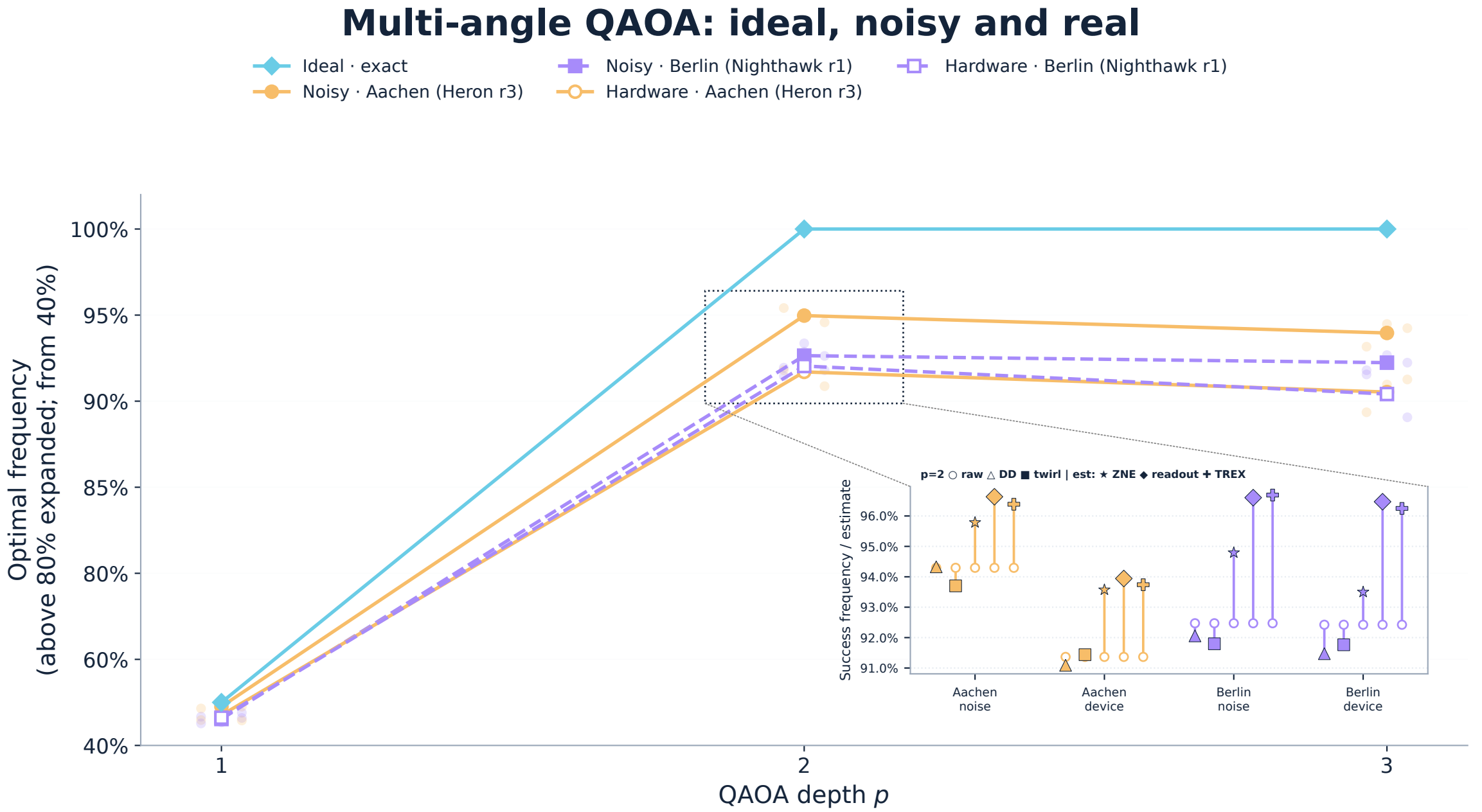

Multi-angle QAOA: ideal, noisy and real
Ideal · exact
Noisy · Aachen (Heron r3)
Noisy · Berlin (Nighthawk r1)
Hardware · Aachen (Heron r3)
Hardware · Berlin (Nighthawk r1)
Optimal frequency
(above 80% expanded; from 40%)
100%
95%
90%
85%
80%
60%
40%
1
2
3
QAOA depth p
p=2 ○ raw △ DD ■ twirl | est: ★ ZNE ◆ readout + TREX
Success frequency / estimate
96.0%
95.0%
94.0%
93.0%
92.0%
91.0%
Aachen noise
Aachen device
Berlin noise
Berlin device